

Tutorial. 7

Q1: Already done in tut. 6

Q2: Let $T \in M_{n \times n}(\mathbb{F})$. Show that T and T^t have same characteristic polynomial, where T^t is the transpose of T .

Solⁿ: Since, $\text{char } T = \det(T - \lambda I)$ where λ is scalar and I is identity matrix

We know $\det(A) = \det(A^t)$

then, $\text{char } T = \det((T - \lambda I)^t)$

$\text{char } T = \det(T^t - \lambda I)$

$\text{char } T = \text{char } T^t$

Q3: Let $T \in M_{n \times n}(\mathbb{F})$. Suppose that T has n distinct eigenvalues $\{a_1, a_2, \dots, a_n\}$

(i) If $|a_i| < 1, \forall i=1, 2, \dots, n$, show that $T_h^m \rightarrow 0$ as $m \rightarrow \infty$ for all $h \in \mathbb{F}^n$

(ii) If $|a_i| > 1, \forall i=1, 2, \dots, n$, show that $T_h^m \rightarrow \infty$ as $m \rightarrow \infty$ for all $h \neq 0 \in \mathbb{F}^n$. Here $T_h^m \rightarrow \infty$ means that some component of T_h^m diverges to ∞ .

Solⁿ: Since T has n distinct eigenvalues that T is diagonalizable.

then $\exists$ an invertible matrix P s.t.

$$T = P D P^{-1} \text{ where } D \text{ is diagonal matrix}$$

with diagonal entries $\lambda_1, \lambda_2, \dots, \lambda_n$ where $\lambda_i, i=1, 2, \dots, n$ are the n -distinct eigenvalues of T .

(i) if $|\lambda_i| < 1 \quad \forall i=1, 2, \dots, n$

$$\text{then } T^m = (P D P^{-1})^m = P D^m P^{-1}$$

$$D^m = \text{diag}(\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m)$$

Since $|\lambda_i| < 1$, so $|\lambda_i|^m \rightarrow 0$ as $m \rightarrow \infty \quad \forall i=1, 2, \dots, n$

$$\text{So, } D^m \rightarrow 0 \text{ as } m \rightarrow \infty$$

Therefore $T^m \rightarrow 0$ as $m \rightarrow \infty$.

(ii) if $|\lambda_i| > 1 \quad \forall i=1, 2, \dots, n$

$$\text{then } T^m = (P D P^{-1})^m = P D^m P^{-1}$$

$$D^m = \text{diag}(\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m)$$

Since $|\lambda_i| > 1$, so $|\lambda_i|^m \rightarrow \infty$ as $m \rightarrow \infty \quad \forall i=1, 2, \dots, n$

$$\text{So } D^m \rightarrow \infty \text{ as } m \rightarrow \infty$$

Therefore $T^m \rightarrow \infty$ as $m \rightarrow \infty$

✶

Q4: Let $C(\mathbb{R})$ be the space of all real valued functions on $\mathbb{R}$. Let $T: C(\mathbb{R}) \rightarrow C(\mathbb{R})$ be the linear operator defined by

$$(Tf)(x) = \int_0^x f(t) dt$$

Show that T has no eigenvalues.

Solⁿ: Let if possible T has an eigenvalue say λ .
Then $\exists$ a function $f \in C(\mathbb{R})$ s.t. $(f \neq 0)$ $Tf = \lambda f$

i.e. $(Tf)(x) = \lambda f(x)$

$$\Rightarrow \int_0^x f(t) dt = \lambda f(x)$$

~~$$\Rightarrow f'(x) = \lambda f(x)$$~~

~~$$\Rightarrow f(x) = A e^{x/\lambda}$$~~ Let $y = f(x)$

~~then $\frac{dy}{dx} = \lambda y$~~

~~Since $f(x) \in C(\mathbb{R})$~~

~~$$\int \frac{dy}{y} = \lambda dx$$~~

~~$$\log y = \lambda x + c$$~~

~~$$y = e^{\lambda x + c}$$~~

~~$$y = c e^{\lambda x}$$~~

~~hence $f(x) = c e^{\lambda x}$~~

diff. both side

$$f(x) = \lambda f'(x)$$

$$f'(x) = \frac{1}{\lambda} f(x)$$

Let $y = f(x)$ then $\frac{dy}{dx} = \frac{1}{\lambda} y$

$$\int \frac{dy}{y} = \int \frac{1}{\lambda} dx$$

$$\log y = \frac{1}{\lambda} x + c$$

$$y = ce^{x/\lambda}$$

$$y = ce^{x/\lambda}$$

So, $f(x) = ce^{x/\lambda}$

Checking consistency

$$(Tf)(x) = \lambda f(x)$$

$$\int_0^x f(t) dt = \lambda f(x)$$

$$\int_0^x ce^{t/\lambda} dt = \lambda ce^{x/\lambda}$$

$$(\lambda ce^{t/\lambda})_0^x = \lambda ce^{x/\lambda}$$

$$(\lambda ce^{x/\lambda} - \lambda c) = \lambda ce^{x/\lambda}$$

$$\Rightarrow \lambda c = 0$$

case (i) $c = 0$ then $f(x) = 0$ which is absurd.

case (ii) $\lambda = 0, c \neq 0$ then $f(x) = 0$ $(Tf)(x) = \lambda f(x) = 0$

$$\int_0^x f(t) dt = 0$$

$f(x) = 0$ which is absurd

hence our assumption is wrong

Q

Q5: Let $A, B \in M_{n \times n}(\mathbb{F})$. Show that if $(I - AB)$ is invertible, then $(I - BA)$ is invertible and

$$(I - BA)^{-1} = I + B(I - AB)^{-1}A.$$

Show that AB and BA have same eigenvalue.

Soln: Since $(I - AB)$ is invertible so $(I - AB)^{-1}$ exist.

$$\text{and } (I - AB)(I - AB)^{-1} = I \Rightarrow (I - AB)^{-1} - (I - AB)^{-1}AB = I \\ \Rightarrow (I - AB)^{-1}AB = (I - AB)^{-1} - I \quad \text{--- (i)}$$

Now consider $(I - BA)(I + B(I - AB)^{-1}A)$.

$$\begin{aligned} &= \cancel{I + B(I - AB)^{-1}A - BA - BAB(I - AB)^{-1}A} \\ &= I(I - BA) + B(I - AB)^{-1}A(I - BA) \\ &= (I - BA) + B(I - AB)^{-1}(A - ABA) \\ &= (I - BA) + B(I - AB)^{-1}A - B(I - AB)^{-1}ABA \end{aligned}$$

So by eqn (i)

$$\begin{aligned} &= (I - BA) + B(I - AB)^{-1}A - B[(I - AB)^{-1} - I]A \\ &= (I - BA) + B(I - AB)^{-1}A - B(I - AB)^{-1}A + BA \\ &= I - BA + BA \\ &= I \end{aligned}$$

which proves $(I - BA)$ is invertible and

$$(I - BA)^{-1} = I + B(I - AB)^{-1}A.$$

Now To show: AB and BA have same eigenvalue

$(\Rightarrow)$ Assume that λ is an e.v. of AB . then $\exists v \neq 0$ s.t.
 $(AB)v = \lambda v$.

Let $w = Bv$

Case (i) $w \neq 0$, since $ABv = \lambda v$
 $BABv = B\lambda v$
 $(BA)w = \lambda(Bv)$
 $(BA)w = \lambda w$
hence λ is an e.v. of BA .

Case (ii) $w = 0$, then $ABv = \lambda v = 0$
since $v \neq 0$ so $\lambda = 0$
 BA must have 0 as an eigenvalue.

$(\Leftarrow)$ Assume that λ is an e.v. of BA . then $\exists v \neq 0$ s.t.
 $BAv = \lambda v$

Let $w = Av$

Case (i) $w \neq 0$ Since $BAv = \lambda v$
 $ABAv = \lambda Av$
 $(AB)w = \lambda w$
hence w is an e.v. of AB

Case (ii) $w = 0$ then $BAv = \lambda v = 0$
Since $v \neq 0$ so $\lambda = 0$
 AB must have 0 as an e.v.